

Supplement operation manual of RHEED monitoring software for RHEED image processing (Scanning functions)



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Ver. 5.1 3'rd Nov. 2009

For PCI-1410, with Scanning

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This manual assumes the users know well about the operation of *Windows* system.

1. Outline

This document is the supplementary manual of the “Operation manual for RHEED Monitoring software”, describing the Scanning function. Refer also the Operation Manual of the above.

2. Working Environment

For the Scanning function, following environments are additionally required with the basic RHHED environment.

- Analog CCD camera (Monochrome); *e.g.* SONY XC-75 + DC700(Power Supply)
(with Camera cable, Lens, *etc.*)
- RHEED Coil Driver (H8 Micro Processor communication unit and cables)
- A Serial port on PC
(or USB Serial converter and USB port)
- RHEED electron Gun with double coil unit

3. Operations

3-1 Selection of the Scanning Mode

Clicking [Video→Video source] on the Menu bar in the Main window shows the Video Mode Selection windows as the next Fig. 1.



(A) Non Scanning Mode

(B) Scanning Mode

(C) Simulation Mode

Fig. 1 Video Mode Selection window

Click “Camera Type” **Camera Type** **A free-running camera** button to show the menu and select “Synchronized camera” from the menu. (Video Mode goes to (b) Scanning Mode automatically)

3-2 In case the Error Message “No Video Signal” shows on the screen

In case that dotted line including the scanning information does not locate properly on the window as shown in Fig. 2.



Fig. 2 “No Video Signal”

In such a case, adjust the dot line position as the followings;

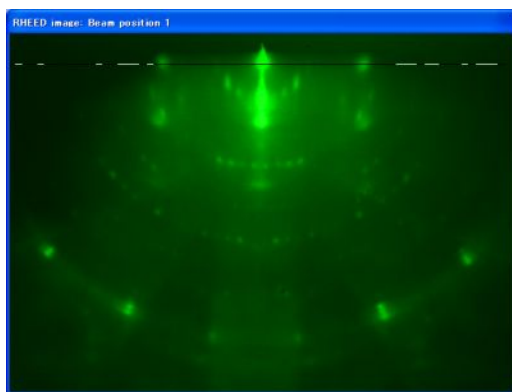
(1) Adjust the Left side offset of the window

The “Video Offset” adjusting tool and the Offset monitor “IDbits”  appear at the left-lower corner of the Video Input Selection window. Adjust the value of “Video Offset” to obtain 4 or 5 of the Offset monitor value (IDbits).

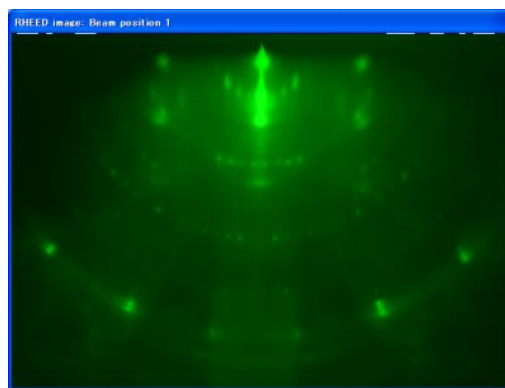
After adjusting the Video Offset, the “Camera Type” should be once set to the “A free-running camera”, to enable the adjustment. Then repeat the procedures of the adjustment of the Video Offset and change the Camera Type” to confirm the result for several times.

(2) Adjustment of the display position of the Dot line information

The dotted line is indicated in the Video Monitoring window, the line must be located at the top of the window as shown in Fig. 3(b), in some cases, it might be positioned in the window(Fig. 3(a)), or disappears to the out of window area.



(a) Dotted line locates in the window



(b) Dotted line locates at the top

Fig. 3 The Dotted line in the Video Monitoring window

Use the “ID line” setting tool  at the center of the window, and place the Dotted line at the top of the window.

After adjusting the ID line, the “Camera Type” should be once set to the “A free-running camera”, to enable the adjustment. Then repeat the procedures of the adjustment of the ID line and change the Camera Type” to confirm the result for several times.

When the Dotted line is properly displayed at the top of the window, the RHEED image will be displayed in the Video Monitoring window. The default value of the ID line is “21”.

3-3 The Main window in the Scanning Mode

In the Scanning Mode by setting the “Camera Type” to “Synchronized camera” in the Video Input Selection window, the Main windows shows the Beam Position area at the bottom of the window as shown in Fig. 4.



Fig. 4 the Main window in the Scanning Mode

The Beam Position area has items of the Beam Table, the Beam Positions, the Beam Control Dial, and the Scan Button. Details will be described in the next section.

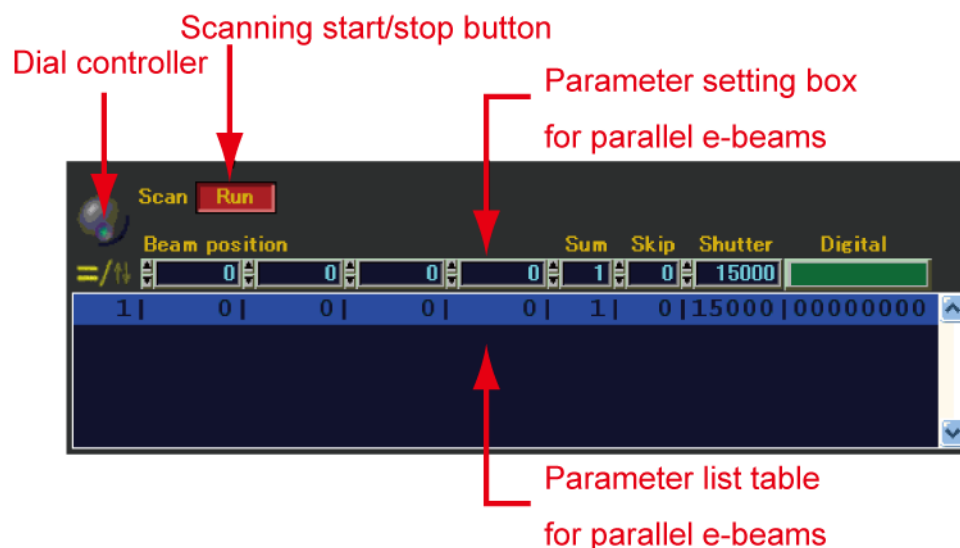


Fig. 5 The Beam Position area

3-4 The Beam Table

A table summarizing the data lines that specifies all the Beam Positions at the lower half of the Beam Position area is called the "Beam Table". User can describe any Beam Position in the Beam Table. The first line shows the default one. The Beam Table can be saved and restored as a data file, which are called the "Setup File".

The "Setup File" can be operated by the following commands in the [File] tag on the Menu bar in the Main window.

[Load Setup] : Reads the Setup File and renews the Beam Table.

[Save Setup] : Over-writes the data file y the current Beam Table.

[Save Setup As] : Saves the current data in the Beam Table into the named file.

3-5 Setup of Beam Table lines

The operation [add, copy, or delete, etc] to each Beam Position line in the Beam Table uses the following commands selectable in the [Beam] menu bar in the Main window.

[Position controls] : Displays the dial for the Beam control.

Same as cricking the Beam Control Dial button  at the upper-left corner in the Beam Position area. Details are described in the section 3-6 "Setup of each Beam Table line".

[Delete all] : Clears all Beam Table (only a blank line remains).

[Add new] : Adds a new line at the next to current Beam Position line.

[Cut] : Deletes the current Beam Position line.

[Copy] : Copies the current Beam Position line.

[Duplicate] : Duplicates the current Beam Position line to the next.

[Paste] : Pastes the copied line.

Multiple lines can be selected (indicated in blue characters) by cricking with the Shift key pressed. It would be convenient in Cutting or Copies many lines at once.

3-6 Setup of each Beam Table line

Values of items in the Beam Position setting tool must be set to specify the beam position or the data gathering frame. Items of the Beam Position Setting tool are as follows shown in Fig. 6.

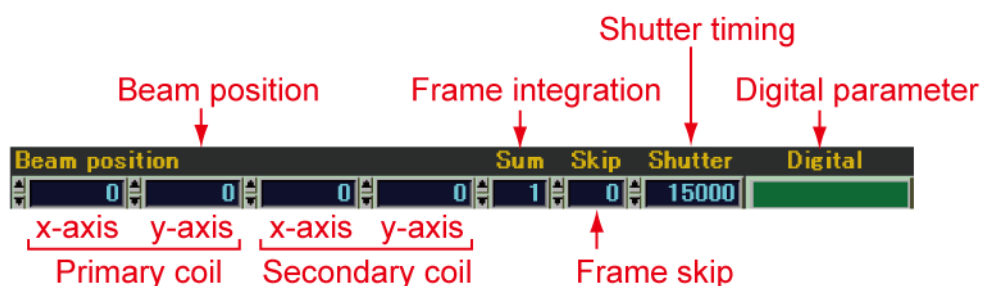


Fig. 6 Beam Position Setting tool

(1) Setup of the Beam Position :

The primary coil (far from the chamber) and the secondary coil (near to the chamber) of the RHEED should be adjusted in the directions of X and Y axes.

There are two ways for adjusting.

(1) Sets the numerical values directly into the Beam Position setting tool

(2) Specifies by the Beam Control Dial

Followings are how to specify them with Beam Position setting tool.

Cricking the Beam Control Dial  at the upper-left corner displays the Beam Control Dials



on the screen. (as described in the section 3-5, cricking [Beam→Position controls] in the Main window shows them). The four dials make line in the same order as the Beam Position Setting tool.

The dial has functions of the fine and rough adjusting.

Fine : touch the small circular area colored in green.

Rough : touch the large circular area behind the green one, colored in purple.

Dragging to turn each dial by the mouse can change the corresponding coil current to adjust the beam position.

Also, clicking the icon  beneath the Beam Control Dials selects the absolute position or the relative position.

Absolute :  (The left side is bright. Enter the absolute position value.)

Relative :  (The right side is bright. Moves by set value from the last position)

Details of the setup of the Beam Position is described in the section 3-7 "Settling the values from the Beam Position".

(2) Frame addition :

This system is acquiring the image data with the frame rate of 30fps (frames per second). (It can switch the beam in every 1/30 sec.)

In case the intensity of the image on the screen is low (dim) or high (bright), the switch timing of the beam can be changed with unit of the frame timing. For example, "Sum=2" makes the beam switch timing to 2 frames (1/15 sec), the intensity of the beam will be brighter. The default value is "1"(no addition).

(3) Frame Skipping :

The beam scanning is performed following the beam position data described in lines of the position data in the Beam Table, in the order of upper to lower. If the Frame Skipping is specified, the beam position line will be vaulted over for the specified number. For example, "Skip=1" moves the beam once in 2 scans (in this case, the data is acquired with the frame rate of 15fps).

When the characteristics of the fluorescent screen shows a rather long afterglow, and shows a clear locus, the Frame Skipping is very effective in reducing it. The default value is "0"(No Skipping).

(3) Shutter Timing :

The shutter opening time can be changed between 97 and 15000 [uS]. The default is 15000 [uS]. In case the intensity is too high, a shorter shutter time avoids the brightness saturation.

(4) Bit Output :

Currently not available. (Reserved)

3-7 Settling the values for the Beam Position

The numerical values of the Beam Position Setting tool for the primary and the secondary coils varied between -2048 and +2047, meaning of this value depends on the geometrical layout of the components, such as RHEED gun coils (the primary, the secondary), the substrate and the fluorescent screen.

Therefore, there need the calculations of the angular value or pixels from the layout of the components.

Actual procedures are as follows;

- (1) Throw the RHEED beam directly onto the fluorescent screen (See the direct beam).
- (2) Find the setup value from the angle and pixel.

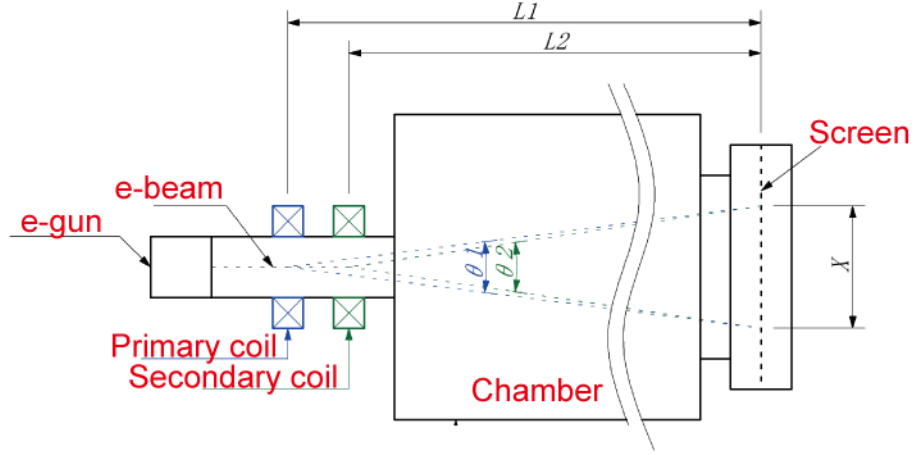


Fig. 7 Schematic Diagram of the RHEED system

As shown in Fig. 7, the distance between the primary coil and the fluorescent screen ($L1$), also that from the secondary coil ($L2$) are already known from an actual survey.

Set the zero value into both X and Y of the secondary coil, and change the X and Y value of the primary coil. Measure the change of the position of the beam on the fluorescent screen by a ruler.

While Fig. 7 shows the case that the Y -axis was changed, measure the variation for X position in the same way. Then find the relation between the setup value for "Beam Position" and angle ($\theta 1$) from the positional change X and the distance $L1$. At the same time, find the beam position from the window and the relation between the setup value and the pixel value.

Next, as same as the above, set the zero value into both X and Y of the Primary coil, and change the X and Y value of the secondary one. Measure the change of the position of the beam and find the relations among the setup values and the angle ($\theta 2$) and pixels.

- (3) Find the relation between coils and the substrate.

Once the relation among the setup value for the "Beam Position" and the beam angle are found, the incident angle (position) of the beam to the substrate can be specified. Figure 8 shows the layout of the coil and the substrate.

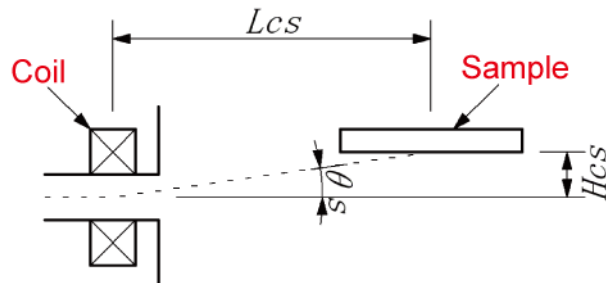


Fig. 8 Schematic layout of the Coil and the Substrate

As same as the section (2), confirm the layout of the coils and substrate from the drawing of the system. Figure 8 simplified the layout, find the two relations of distances for coils. From these relations, you can find how to control the beam position by the coil current.

From Fig. 8, an incident at the center of the substrate from the center of the coil is calculated with the vertical and horizontal distance between the coil center and the substrate of H_{cs} and L_{cs} , respectively, from the following equation (1).

$$\theta_s = \tan^{-1}(H_{cs}/L_{cs}) \quad -(1)$$

Since the relation between the angle and the beam position has been already found in the above subsection (2), the corresponding angle of (θ_s) can be specified to the Beam Position Setup tool. While the angle depends on the substrate height, conditions of the substrate holder, or an offset of the coil current, approximate value of the incident position can be specified and the irradiated position can be confirmed by the naked eyes, by the electro-luminescence from the sapphire substrate.

(4) The primary and the secondary coils

By the combination of the primary coil and the secondary coil, the followings become operationable;

- Change of the e-beam incident angle : specified Y scanning axis
- Parallel beam incidence to the substrate : specified X scanning axis

Figure 9 shows the schematic illustration of the beam trace with the double coil scanning. It shows how to change the incidence angle by the Y-axes scanning of the primary and the secondary coils.

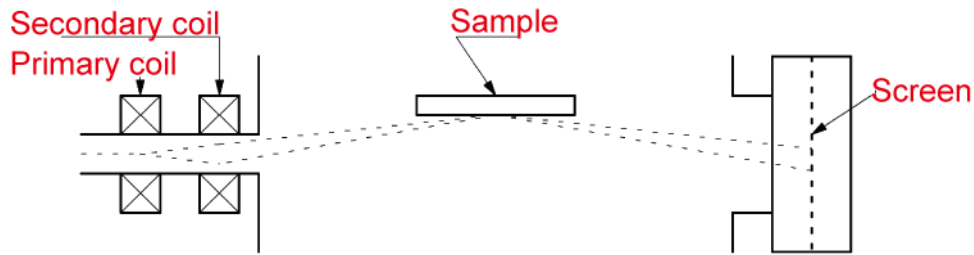


Fig. 9 Schematic illustration of the e-beam trace with the double coils

With this configuration, when a shallow (parallel to the substrate) incident angle is desired, the e-beam should be scanned upper direction by the primary coil, then scanned lower direction by the second coil. On the contrary, for the deep (vertically) incident angle, the combination of the lower direction by the primary coil and the upper by the second one realizes it.

For the X direction, when the parallel e-beam to the substrate is desired to irradiate the left part of the substrate, from the view point of the RHEED gun, the e-beam should be scanned to left by the primary coil, then scanned to right by the secondary one, and vice versa. For the values corresponding to the scanning angle, values found in the subsection (2) can be used.

(5) Preparing the Beam Table

The Beam Table is accomplished by adding each Beam Position line including the every Beam Position data specified by the procedure described in subsection (4). Therefore, this process is the most important for preparing the Beam Table.

This Beam Table can be stored as the Setup file, and be used repeatedly, or re-used with a small modification (Details are described in the section 3-4 “Beam Table”).

(6) Some notices in setting the Beam Position

The hysteresis of the coil unit may affect the beam position with the moving direction of the beam. Therefore, in preparing the Beam Position, the different beam scanning direction may show the different beam position actually, even for the same Beam Position value. Fluctuation can be avoided by making a dummy point in the Beam Table line to ensure the same beam moving direction as at the setting.

3-8 Start and Stop of the Beam Scanning

Start the beam scanning by clicking “Run” on the “Scan”  button. The e-beam will be scanned according to the beam position data described in the first Beam Table line to the last line in turn.

When the “Scan” button  is “Hold”, the beam position value at the currently active beam table line (indicated in blue) is outputted. Click the “Scan” button to manipulate the scanning.

3-9 Supplementary setups of the Video Monitoring

Please refer the “Operation manual for the RHEED monitoring program”, details on the setup of the Video Monitoring are described in the section 3-12 of the manual.

Clicking [Video→Display settings] on the Menu bar in the Main window opens the Video Monitoring window. In the Scanning Mode, select whether or not to use the content of the Beam Table as the parameters.

In the “Multibeam scaling”  selector, select one of the followings.

“Absolute”: Not use the Beam Table setups.

“Per beam” : Uses the Beam Table as the parameters.

When the Beam Table setup used as the scanning parameters, the beam number in the Beam Table will be added into the frame number , in order to obtain a brighter (clear) images.

4. Troubleshooting

No.	Trouble Matter	Causes and Solutions
1	The e-Beam scanning is differ from the specified value (X/Y axes, moving direction).	The connection to the RHEED gun coils might be changed. When the coil configuration was changed after the adjustment of the coil unit, the inner connection of the coils should be changed. In such a case, contact the machine maker or the machine vender.
2	No image was displayed and shown the error message“No Video Signal”. (H8 lamp is Green in the Main window)	The dotted line including the scanning information does not locate the proper position. Refer the section 3-2, and adjust the dotted line at the proper position.
3	No image was displayed and shown the error message“No Video Signal”. (H8 lamp is Red in the Main window)	The communication with the H8 coil driver board might not work well. Check the followings. [1] Check the Cable and Connectors. Is The cable between the PC and the coil driver board connected firmly? Is there any inner connection wire snapped down, or the connector come off? [2] Check the Port Cricking [Video→Video source] from the Menu bar shows the Video Input Selection window, check if the setup value Serial port ASRL1 at the center of the Scanning window matches the actually used port number. If not, set the correct port number. In case that there is no above issue, contact the machine maker or the machine vender.
4	The Beam position is fluctuated even fro the same Beam Table number.	The hysteresis of the coil unit may affect the beam position with the moving direction of the beam. Check the beam moving direction is same as at the setting. Or make a dummy position to ensure the correct moving direction. Refer details described in the section 3-7 “Settling the values fro the Beam Position”.